CP–7990

Sub. Code
11

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

ENGLISH AND COMMUNICATION SKILLS

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) Read the passage carefully and answer the questions given below : (8)

Five roommates Randy, Sally, Terry, Uma, and Vernon each do one housekeeping task of mopping, sweeping, laundry, vacuuming, or dusting one day a week, Monday through Friday.

* Vernon does not vacuum and does not do his task on Tuesday.
* Sally does the dusting, and does not do it on Monday or Friday.
* The mopping is done on Thursday.
* Terry does his task, which is not vacuuming, on Wednesday.
* The laundry is done on Friday, and not by Uma.
* Randy does his task on Monday.

(i) What task does Terry do on Wednesday?

(ii) What day does Uma do her task?

(iii) What task does Vernon do?

(iv) What day is the vacuuming done?

(b) Write an essay on demonetization and its impact. (7)

Or

(c) Write a précis of the following passage : (8)

A life of action and danger moderates the dread of death. It not only gives us fortitude to bear pain, but also teaches us at every step the precarious tenure on which we hold our present being. Sedentary and studious men are the most apprehensive on this score. Dr. Johnson was an instance in point. A few years seemed to him soon over, compared with those sweeping contemplations on time and infinity with which he had been used to pose himself. In the still life of a man of letters there was no obvious reason for a change. He might sit in an arm chair and pour out cups of tea to all eternity would it had been possible for him to do so. The most rational cure after all for the inordinate fear of death is to set a just value on life. If we mere wish to continue on the scene to indulge our head-strong humour and tormenting passions, we had better be gone at once,; and if we only cherish a fondness for existence according to the good we desire from it, the pang we feel at parting which it will not be very severe.

(d) What are the characteristic features of a good essay? (7)

2. (a) What are the basic skills required to build written communication skills? (8)

(b) Write a letter in response to an advertisement : (7)

Wanted Accountant. B. Com ., with computer knowledge Preference will be given to people with experience. Apply within 7 days to The Personal Manager East West Trading Company, 82, Usman Road, T. Nagar, Chennai-17.

Or

(c) Write the parts of letter in detail. (8)

(d) Write a letter to the Municipal Commissioner about the bad condition of street/road. (7)

3. (a) How will you prepare a factual report? (8)

(b) Write a letter to a daily newspaper about nuisance of beggars in trains. (7)

Or

(c) What are the basic steps one need to know while reporting accidents in daily newspaper? (8)

(d) Write a letter to your younger brother advising him to read good books. (7)

4. (a) What is communication? What are the barriers encountered by a communicator?

(b) What are the strategies you will follow to develop the speaking skills?

Or

(c) What are the differences between verbal and non-verbal communication. (8)

(d) What are the types of communication? Elaborate in detail. (7)

5. (a) What are the benefits of getting speech training? (8)

(b) What are the qualities of a good debater? (7)

Or

(c) Write a welcome speech for the chief guest attending your sports day function. (8)

(d) What are the qualities that an interviewer would expect from an interviewee? (7)

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CP–7991

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAUTICAL MATHEMATICS — I

(Upto 2015 batch)

Time : 3 Hours Maximum : 100 Marks

Answer **all** questions. (5 × 20 = 100)

1. (a) (i) Obtain the equations of two lines of regression for the following data : Also obtain the estimate of X for $Y = 70$. (15)

X :	65	66	67	67	68	69	70	72
Y :	67	68	65	68	72	72	69	71

(ii) State and prove Baye's theorem. (5)

Or

(b) (i) If $\vec{a} = \vec{i} - 2\vec{j} + 3\vec{k}$, $\vec{b} = -2\vec{i} + 3\vec{j} - 4\vec{k}$ and $\vec{i} = -\vec{j} + 2\vec{k}$, then find $\vec{a}\cdot(\vec{b}\cdot\vec{c})$, $\vec{b}\cdot(\vec{c}\cdot\vec{a})$, $\vec{a}\cdot(\vec{b}\times\vec{c})$, $\vec{b}\cdot(\vec{a}\times\vec{c})$ and $\vec{a}\times(\vec{b}\times\vec{c})$. (15)

(ii) Show that the following vectors are coplanar $\vec{i} - 2\vec{j} + 3\vec{k}$, $-2\vec{i} + 3\vec{j} - 4\vec{k}$ and $-\vec{j} + 2\vec{k}$. (5)

2. (a) (i) Find the equation of the circle passing through the points $(0, 1)$, $(2, 3)$ and $(-2, 5)$. (10)

(ii) Find the equations of axes and length of axes of the ellipse $6x^2 + 9y^2 + 12x - 36y - 12 = 0$. (10)

Or

(b) (i) A satellite is travelling around the earth in an elliptial orbit having the earth at a focus and eccentricity 1/2. The shortest distance that the satellite gets to the earth is 40 kms. Find the largest distance that the satellite gets from the earth. (10)

(ii) Find the centroid of the quarter circle $x^2 + y^2 = r^2$, $0 \le x \le r$. (10)

3. (a) (i) If ares be drawn from the angles of a spherical triangle to meet the mid-points of the opposite sides, and if α, β be the parts of are which bisects the sides α, the show that $\frac{\sin\alpha}{\sin\beta} = 2\cos\frac{1}{2}\alpha$. (10)

(ii) In an equilateral spherical triangle ABC, prove that $1 - 2\cos A = \tan^2\frac{1}{2}a$. (10)

Or

(b) (i) Solve the triangle when $A = 115°20'$, $C = 80°\,57'$ and $C = 90°$. (10)

(ii) Define right-angled spherical triangle and write any five of its properties. (10)

4. (a) (i) If $y = \tan^{-1}\left(\frac{2x}{1-x^2}\right) + \tan^{-1}\left(\frac{3x-x^2}{1-3x^2}\right) - \tan^{-1}\left(\frac{4x-4x^3}{1-6x^2+x^4}\right)$, show that $\frac{dy}{dx} = \frac{1}{1+x^2}$. (10)

(ii) If $x(1+y)^{1/2} + y(1+x)^{1/2} = 0$ prove that $\frac{dy}{dx} = -\frac{1}{(1+x)^2}$. (10)

Or

(b) (i) Evaluate $\int \frac{x^{24}}{x^{10}+1}\, dx$. (10)

(ii) Evaluate $\int \frac{2dx}{(1-x)\,(1+x^2)}$. (10)

5. (a) (i) Test for the consistency of the following systems of equations and solve, if consistent.

$x-3y-8z=10,\ 3x+y-4z=0,\ \ 2x+5y+6z=13$. (10)

(ii) Find the rank of the matrix $A=\begin{bmatrix} 2 & -2 & 1 \\ 1 & 4 & -1 \\ 4 & 6 & -3 \end{bmatrix}$. (10)

Or

(b) (i) Find the eigen values and eigen vectors of the matrix. $A=\begin{bmatrix} 11 & -4 & -7 \\ 7 & -2 & -5 \\ 10 & -4 & -6 \end{bmatrix}$. (10)

(ii) Verify Cayley-Hamilton theorem for the matrix. $A=\begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$. (10)

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAUTICAL PHYSICS AND ELECTRONICS — I

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) State Kepler's law. (3)

   (b) Briefly explain the planetary motion and centre of gravity. (12)

   Or

   (c) Define impulse, impact. (4)

   (d) State and explain the Lami's theorem. (5)

   (e) Explain the direct and oblique impact. (6)

2. (a) Define battery and their types. (7)

   (b) State and explain the photo electric effect and photo voltaic cell. (8)

   Or

   (c) Draw and explain the hysterises loop and also for magnetic elements of the earth. (7)

   (d) Define :

      (i) RMS

      (ii) Peak value

      (iii) Phase value

      (iv) Solenoid. (8)

3. (a) Briefly explain the Bernoulli's equation and its applications. (7)

(b) Describe Poiseuille's method to determine the co-efficient of viscosity of a liquid. (8)

Or

(c) State and explain the Stoke's law. (7)

(d) Describe the Bourdon pressure gauge to determine the viscosity of a liquid. (8)

4. (a) Draw and describe the working of a photo diode. (5)

(b) Explain the extrinsic semiconductor with some examples. (5)

(c) State and derive the Einstein equation. (5)

Or

(d) Write a short note on : (10)

(i) LED

(ii) LDR

(iii) Photo voltaic cell

(iv) Photo electric effect

(v) Luminous intensity.

(e) State and explain the law of illumination. (5)

5. (a) Draw and explain V-I characteristics of junction diode. (7)

(b) Describe the construction and working of a full wave bridge rectifier and how the bridge rectifier rectifies with filter circuit explain it. (8)

Or

(c) What is diode? (2)

(d) Explain the working of a diode during forward and reverse bias. (8)

(e) Briefly explain how the zener diode be used as voltage stabilizer with necessary circuit. (5)

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAVIGATION – I

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Part A (5 × 15 = 75)

Answer **all** questions.

1. (a) A vessel in lat 69º 11'N 099º 39'W steers a course of 270º for a distance of 794.8 miles. Find the position arrived.

Or

(b) Find the course and distance from A 04º 16'S 177º 37'W to B 02º 29'N 179º 24'E.

2. (a) With the help of diagrams, define :

(i) LHA

(ii) GHA

(iii) Mercator sailing formula

(iv) Deviation

(v) Variation.

Or

(b) Compare between plane sailing and mercator sailing.

3. (a) Find the GC course and distance from A 24° 00'N 074° 15'W to B 46° 00'N 053° 45'W.

Or

(b) What is total solar eclipse? (10)

(c) What is vertex of a GC? (5)

4. (a) Define Keplers first law and second law of planetary motion.

Or

(b) Define and write short notes on :

(i) Inferior planets

(ii) Superior planets

(iii) Conjunction

(iv) Opposition

(v) Perigee

(vi) Apogee.

5. (a) At noon on 20 July, your ship was in position 32° 43.8'N 017° 26.8'W steering 219° (c) Dev 3°E Var 18°W. Your ship maintained this course at a steady speed of 15 knots till noon next day. Clocks were retarded for total one hour. Find the DR position of your ship at 1200 hrs on 21 July. If the fix at noon was 27° 24.3'N 020° 01.0'W find the set and rate of current.

Or

(b) Draw the suitable figure and box the magnetic compass by points. How many degrees is one point?

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

SHIP OPERATION TECHNOLOGY – I

(upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) Write short notes for the following : (15)

   (i) Freighter

   (ii) Bulk carrier

   (iii) Lumber carrier

   (iv) Car carrier

   (v) Gas carrier.

Or

(b) Draw a neat diagram of poop deck and name all the parts for mooring arrangements. (7)

(c) Describe the following terms : (8)

   (i) Midship

   (ii) Stem and stern

   (iii) Bow and Quarter

   (iv) Hatch cover and coamings.

2. (a) What is magnetic compass? (2)

(b) Make a neat diagram of a magnetic compass card and mark all points on the card. (10)

(c) How do you read the point when you are on bridge watch? State the degrees for 1 point. (3)

Or

(d) Draw a neat diagram and describe the uses of the following tanks.

(i) DB tanks (3)

(ii) TST tanks (3)

(iii) Fore peak and Aft Peak tanks. (4)

(e) Explain the preparation for painting on a metal surface in a ship. (5)

3. (a) What is lead line? What are the markings you do to take soundings with the help of Lead line? (8)

(b) Explain different types of bottom logs. (7)

Or

(c) What is Look-out? (2)

(d) What are the duties and procedure for reporting in a bridge as a look-out man? (5)

(e) What are the emergencies to post extra look-out and mention the places to post look-out at security watch level-2? (6)

(f) What is access point? (2)

4. (a) What is PPE? List all protective equipments and describe each item on them. (10)

(b) Why should wear the safty gears when working on deck? Explain any one incidents non-wearing of safty gears. (5)

Or

(c) What are the types of fiber ropes? Describe. (5)

(d) What are the materials used for making a fiber rope? (5)

(e) Explain the types of lay and advantages of each type of fibre ropes. (5)

5. (a) Draw a neat diagram of tackle and name the parts of tackle. (3)

(b) What are the advantages and disadvantages when using a tackle? (7)

(c) What is union purchase? (2)

(d) What are the relation between angle and runner on a derrick? (3)

Or

(e) What are the types of slings using for cargo operation in a ship? Explain. (6)

(f) What do you mean by the term "Battening down". (2)

(g) What are the precautions to be taken to prevent an accident while cargo operations? (7)

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B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

VOYAGE PLANNING AND COLLISION PREVENTION AND MARINE COMMUNICATION – I

(Upto 2015 Batch)

Time : 2 Hours Maximum : 25 Marks

Answer **all** questions. (5 × 5 = 25)

1. (a) Explain compass error with a neat diagram and an example.

Or

(b) List all the rules which may applicable in sight of one another.

2. (a) Name two types of charts with respect to the projections. List the advantage and disadvantage of Mercator charts.

Or

(b) Differentiate small scale and medium scale charts and also name the categories of chart.

3. (a) What is meant by chart datum? If height of the tide is given as 3.5 m and charted depth is as marked as 4.5 m, what is the actual depth at the mark?

Or

(b) A light house is marked on a chart with "Fl(2) 12s 38m 27M". What do you understand by this?

4. (a) Explain magnetic compass and its location on board a ship. Why magnetic compass is standard compass.

Or

(b) Explain gyro compass and how it differs from magnetic compass. Also what is gyro error?

5. (a) Explain briefly :

(i) Underway

(ii) Making way through water

(iii) Not Under Command (NUC).

Or

(b) Define :

(i) Power driven vessel

(ii) Vessel at anchor

(iii) Restricted visibility.

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21

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAUTICAL MATHEMATICS — II

(Upto 2015 batch)

Time : 3 Hours Maximum : 100 Marks

Answer any **five** questions.

$(5 \times 20 = 100)$

1. (a) Explain the geometric representation of $\frac{z_1}{z_2}$. (10)

   (b) State and prove Demoivre's theorem. (10)

2. (a) If $2\cos\theta = x + \frac{1}{x}$, prove that

   (i) $2\cos r\theta = x^r + \frac{1}{x^r}$,

   (ii) $\frac{x^{2n}+1}{x^{2n-1}+x} = \frac{\cos n\theta}{\cos(n-1)\theta}$. (10)

   (b) Separate $\sin^{-1}(\cos\theta + i\sin\theta)$ into real and imaginary parts, where θ is a positive acute angle. (10)

3. (a) Find the first and second derivative of $f(x)$ at $x = 1.5$. (10)

x	1.5	2.0	2.5	3.0	3.5	4.0
$f(x)$	3.375	7.0	13.625	24.0	38.875	59

(b) Evaluate $\int_0^1 e^{-x^2} dx$ by dividing the range of integration into four equal parts using trapezoidal rule and Simpson's rule. (10)

4. (a) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using Simpson's $\frac{3}{8}$ rule taking $h = \frac{1}{6}$ and Weddle's rule taking $h = \frac{1}{6}$. (10)

(b) Find the work done by the force $\bar{f} = (2xy + z^3)\bar{i} + x^2\bar{j} + 3xz^2\bar{k}$, when it moves a particle $(1, -2, 1)$ to $(3, 1, 4)$ along any path. (10)

5. (a) Verify Gauss divergence theorem for $\bar{f} = x^2\bar{i} + z\bar{j} + yz\bar{k}$ over the cube bounded by $x = \pm 1, y = \pm 1, z = \pm 1$. (15)

(b) Show that

$$\bar{f} = (2x^2 + 8y^2xz)\bar{i} + (3x^3y - 3xy)\bar{j} - (4y^2z^2 + 2x^3z)\bar{k}$$

is not solenoid. (5)

6. (a) Form the differential equation which has $y = ax^2 + bx$ as a general solution, where a and b are arbitrary constants. (10)

(b) Solve : $\frac{dy}{dx} - y \tan x = e^x \sec x$. (10)

7. (a) Solve

(i) $\tan y \sec^2 x \, dx + \tan x \sec^2 y \, dy = 0$ (5)

(ii) $xdy - ydx = \sqrt{x^2 + y^2} dx$. (5)

(b) The rate at which are substance combines with another is supposed to be proportional to the amount of the first substance remaining. If there be 15 grams of the first substance when $t = 0$ and 5 grams when $t = 8$ seconds, find how much will be left when $t = 5$ seconds. Also find the value of t when there is one gram left. (10)

8. (a) Solve : $\left(D^4 + D^3 + D^2\right)y = 5x^2 + \cos x$. (10)

(b) Solve : $\left(3D^2 + D - 14\right)y = 8e^{2x} + \cos 5x$. (10)

CP–7997

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22

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Nautical Science

NAUTICAL PHYSICS AND ELECTRONICS — II

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

$(5 \times 15 = 75)$

1. (a) Name the parts of nuclear reactor and discuss their functions with safety precautions. (8)

   (b) Discuss the role of satellite for the weather forecast. (7)

   Or

   (c) Explain the electrostatic charging of oil in pipeline flow. (7)

   (d) Discuss various causes for formation of changes in oil and their remedies. (8)

2. (a) Derive an expression for resonant frequency of a series LCR circuit. (8)

   (b) A radio can turn on frequency of 1 MHz. If its LC circuit has an effective inductance or 200 μH. What is capacitance value? (7)

   Or

   (c) Discuss the various factors in the radio wave communication. (8)

   (d) Draw the radiation pattern in Hertz antenna and explain. (7)

3. (a) Subtract $(110011)_2$ from $(1111101)_2$. (3)

(b) Explain the construction and working of a NAND gate with the truth table. (5)

(c) With the neat diagram, explain the function of J-K flip flop. (7)

Or

(d) NAND and NOR are universal gates. Explain with neat diagram. (10)

(e) Explain the working of monostable multivibrator with diagram. (5)

4. (a) Discuss the common base mode configuration of a transistor with its applications. (8)

(b) Explain the operation of transistor on an amplifier. (7)

Or

(c) Transistor acts as switches — Explain. (7)

(d) Discuss the various circuits in which transistor as switches. (8)

5. (a) Discuss the FM modulation with necessary theory.(8)

(b) Explain the function of AM transmitter with neat block diagram. (7)

Or

(c) Discuss the working of superhetrodyne receives with neat block diagram. (9)

(d) Discuss the characteristics of radio receives. (6)

CP–7998

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23

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Nautical Science

NAVIGATION – II

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) Explain and write notes on :
   (i) Semi diameter.
   (ii) Polar gnomonic projection.
   (iii) Prime Vertual.
   (iv) SHA.
   (v) Amplitude.

Or

(b) What are the adjustable errors of a sextant? (7)

(c) Explain with diagram Keplers all three laws a Planetary motion. (8)

2. (a) Explain reasons for unequal daylight and darkness through the year. (8)

(b) Compare mercator and gnomonic charts w.r.t. advantage and disadvantages. (7)

Or

(c) Define Apogee, Perigae, GMT, LMT? (8)

(d) What is "V" correlation? Why it must be applied for planets and moon? (7)

3. (a) Find LMT when chron time was $05^d\ 23^h\ 15^m\ 04^s$ chronometer for $00^m\ 04^s$ fast in longitude 95° 30' E? (5)

(b) Explain three kinds of lunar eclipse. (5)

(c) Explain conjunction. (5)

Or

(d) Write short notes on comets. (5)

(e) What is time zone? How will you check the error of a chronometer? (5)

(f) On 20th January 2008 in DR 54° 24' S 046° 27' W the sunset bearing 234°(c). If variation was 3°w. Find the deviation of compass? (5)

4. (a) On 05th March 2008, AM ship in DR 38° 11's 151°10' E, the sextant attitude of Suns LL was 35° 59.1' E, when chron showed $10^H\ 54^m 54^s$ (error $00^m\ 46^s$ slow). If IE was 1.3' off the arc and HE was 30m. Find the intercept and direction of LOP?

Or

(b) On 22 September 2008, pm at ship in DR 48° 20' N 085° 40' E, the sextant attitude of suns UL was 20° 04.9' when the GPS clock showed $10^H 09^m 38^s$. If IE was 2.2′ on the arc and height of eye was 25m. Find LOP and longitude where it cuts the DR latitude?

5. (a) What is international data line? Why should a vessel advance or retard one day upon crossing IDL? (8)

(b) What is "V" correction? Why it must be applied for planets and moon? (7)

Or

(c) What are the adjustable errors of a sextant? (8)

(d) Explain first point of aries and first point of libra with the help of diagrams. (7)

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24

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

SHIP OPERATION TECHNOLOGY – II

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) Describe the following (15)

   (i) Grain capacity

   (ii) Bale capacity

   (iii) Load Density

   (iv) Cargo density

   (v) Ullage and Soundings.

Or

(b) What is the significance of Loadline? (5)

(c) How are the loadlines important in planning cargo carrying capacity? Explain in detail. (10)

2. (a) Describe the basic needs of ballasting and de-ballasting when loading and discharging cargo. (9)

(b) Describe the following. (6)

(i) Stowage and stawage plan.

(ii) Segragation and separation of cargo.

Or

(c) What do you understand by HRU? Explain the function of an HRU.

(d) What is Life Jacket?

(e) What are the requirement as per solas, various attachments and markings on a life Jackets?

3. (a) A tank of $2400\,m^3$ volume and 12 m depth, has vertical sides and Horizontal bottom, find how many tonnes of oil of RD 0.7. it can hold allowing 2% of the volume of the tank for expansion. State the village on loading. (10)

(b) What is oxidation? How cause for fire? (5)

Or

(c) What do you understand by ISC? Draw a neat diagram with correct dimension and explain its use. (8)

(d) What are the preparation prior Safety Equipment Survey (SEQ) on board a ship. (7)

4. (a) Describe the following. (15)

   (i) Flash point

   (ii) Ignition point

   (iii) Sponteneous combustion

   (iv) Fire Triangle

   (v) Flammable rang

   (vi) Spread of fire by conduction.

Or

   (b) What is muster list? Describe the duties for each personnel on board at Boat and fire drills. (8)

   (c) Why all crew familiarization required with all LSA and FFA equipments on signed on a ship. And how frequently to conduct Boat and Fire drills? (7)

5. (a) List all the Life Boat items to be kept ready at all times. Explain. (9)

   (b) What are the marking of a Lift Boat? (3)

   (c) Which one is determining the capacity of a Life boat? (3)

Or

   (d) Describe the following.

   (i) Prevention against Hypothermia.

   (ii) First and for shock, wounds and fractures.

   (iii) The procedure to land a life boat in Heavy Surf.

   (iv) Donning of Life Jackets.

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25

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

VAYAGE PLANNING, COLLISION PREVENTION AND MARINE COMMUNICATION – II

(Upto 2015 batch)

Time : 2 Hours Maximum : 25 Marks

Answer **all** questions. (5 × 5 = 25)

1. (a) Define the following as rules 21. (2)

   (i) Mast head light.

   (ii) Side lights.

   (b) What is the arc of visibility of navigation light as prescribed in rule No. 22. Vessel greater than 100 m in length? (3)

Or

   (c) State the day and night signals (shape and light) for the following. (5)

   (i) Vessel aground

   (ii) Fishing vessel

   (iii) Vessel carrying dangerous cargo

   (iv) Vessel not under command

   (v) Vessel at anchor vessel length greater than 100 m.

2. (a) What are the publications to carry on board ship? (3)

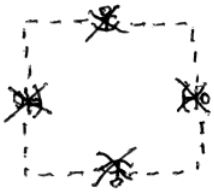
(b) What is the use of ASO? (2)

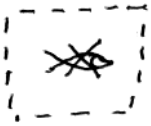
Or

(c) What is notices to mariners? List the available information in it. (3)

(d) What is the uses of Mariner's hand book? (2)

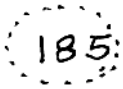
3. (a) Identity the following chart symbols. (5)

(i) 

(ii) 

(iii)

(iv)

(v) 

Or

(b) Define the following : (5)

(i) Quarantine flag

(ii) Ensign staff

(iii) Gaff

(iv) Triadic stay

(v) Jack staff

4. (a) Describe the following : (5)

(i) Land fall ahead of your vessel-action.

(ii) Seen thick fog batch- your action.

(iii) How do you select a suitable anchorage?

Or

(b) Find the height of tide and depth of water at 1430 hrs on March 2nd at a position off Singapore. (Use Tide Chart). (5)

Extracts from ATT	
Zone Time –0800	
Time	Height
0014	2.7 m
0603	0.8 m
1209	2.9 m
1830	0.6 m

5. (a) Define the following : (5)

(i) Radar

(ii) ECDIS

(iii) TSS

(iv) Tidal streams.

Or

(b) What is the difference between Racon and Ramark? (2)

(c) What is chart correction? Describe. (3)

CP–8001

Sub. Code
26

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

METEOROLOGY AND OCEANOGRAPHY - I

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

$(5 \times 15 = 75)$

1. (a) Describe the composition and structure of the atmosphere with suitable diagram.

Or

(b) Explain about the solar energy interaction with the atmosphere.

2. (a) Give a short note on:

(i) Relative Humidity

(ii) Dew point and frost point temperature

(iii) Super cooling.

Or

(b) Describe about the stability analysis of atmosphere.

3. (a) What are the principles and methods of surface meteorological observations?

Or

(b) Describe in detail about the different forms of condensation.

4. (a) Give an account on physical properties of sea water.

Or

(b) Describe the horizontal distribution of salinity of ocean water and the factors affecting the distribution.

5. (a) Explain the use of mercuric barometer and what are the corrections made in reading to enter in log book.

Or

(b) Write a note on psychrometer and wind wane.

CP–8002

Sub. Code
31

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Nautical Science

NAVIGATION — III

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) Find the initial course, final course and distance along a GC track from

30° 20.00's 142° 50'W to 50° 40.0's 170° 30.0'E?

Or

(b) Find the initial course, final course and distance in a great circle track from

06° 00'N 079° 00'W to 38° 00's 179° 00'E?

2. (a) Describe with diagram the working principle of an echo sounder? What is SONAR?

(b) Describe errors of echo sounder.

Or

(c) What are the errors of a ship gyro compass?

(d) Define Tilt, Drift, Precession.

3. (a) Explain with diagram phases of moon.

(b) What is the age of moon?

Or

(c) Define :
(i) Conjunction
(ii) Opposition
(iii) Quadrature
(iv) Comets
(v) Meteors.

4. (a) What is a circumpolar body?

(b) What is the vertex of a GC track? Draw diagram and explain.

Or

(c) What are the conditions necessary for a body to be circumpolar?

(d) Define Umbra, Penumbra, Lunar day.

5. (a) On 05th March 2008 is position 32° 12'N 178° 16'E the rising sun bore 100°(C) of variation was 3°E. Find the deviation of compass.

Or

(b) On 13th September 2008 in DR position 23° 21'S 047°18'W, the observed azimuth of the sun was 046°(C), when chronometer showed 01^{H} 08^{m} 10^{s}. If chron error was 02^{m} 12^{s} slow and variation was 3°W. Find the deviation of the ship compass?

CP–8003	Sub. Code
	32

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

SHIP OPERATION TECHNOLOGY — III

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) What are the preparation for berthing and leaving quays under various conditions of wind and tide?(12)

   (b) What is man overboard? State the man overboard signal (sound signal). (3)

   Or

   (c) What are precautions and checks on Heavy weather conditions? (7)

   (d) What is turning circle? Explain with diagram and mark transfer, advance. (4)

   (e) What is the meaning of following Transfer, Advance and Tactical Diameter? (4)

2. (a) What are the precautions when your own vessel Anchored at Tidal port and in confined water? (7)

   (b) Describe the following terms : (8)

   (i) Types of Mooring

   (ii) Foul Anchor

   (iii) Preparation for Letting go

   (iv) Engage anchor at gear.

   Or

(c) What is anchor? Name the types of anchors. (3)

(d) What are the difference between stock Anchor and Stockless Anchor? (6)

(e) Draw a neat sketch of stockless anchor and mark the name of parts. (6)

3. (a) What are the safety precautions when weighting anchor? (5)

(b) As an officer on watch your own vessel dragging – your action and also nearby vessel dragging – your action. (10)

Or

(c) What do you mean by shallow water effect? (5)

(d) What are the various effects while maneuvering in shallow water and in confined water? Explain in detail. (10)

4. (a) What are the precautions to take in maneuvering for launching of Life Raft? (7)

(b) Explain the procedures for recovery of life raft. (6)

(c) What is HRM? Period of validity. (2)

Or

(d) Draw neat diagram of Navigation lights panel and explain the arc of visibility and characteristics of lights. (10)

(e) List out the safety construction certificates. (5)

5. (a) What are the general procedures for dry docking? (8)

(b) What are the methods and procedures for ship's maintenance and inspection for the DB tanks and fresh water tanks? (7)

Or

(c) Draw a neat sketch of Load line Marks and explain the markings. (10)

(d) What is DWA? How does it calculate? (5)

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CP–8004

Sub. Code
33

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

VOYAGE PLANNING, COLLISION PREVENTION AND MARINE COMMUNICATION – III

(upto 2015 Batch)

Time : 2 Hours Maximum : 25 Marks

Answer **all** questions.

Each question carries 5 marks.

1. (a) Briefly explain the four stages in any passage that a ship undertakes.

Or

(b) (i) Name six publications consulted for passage planning.

(ii) Explain briefly OPW and ASD.

2. (a) (i) Define Meteorological visibility.

(ii) When will an observer on a vessel (height of eye = 49 m) approaching a light (fl 5 sec 121 m 25M) see the light for the first time in normal visibility during night. Will the observer see the light at rising horizon then?

Or

(b) What are the two criteria for determining the maximum range at which a light can be seen and what are the factors it depends on?

3. (a) (i) Write short note on navigational warnings.

(ii) What are the types of radio navigational warnings?

Or

(b) Write in brief, what, information is provided by different volumes of Admiralty list of Radio signals. What are they popularly known as?

4. (a) Write five factors which must be considered in placing the degree of reliability on a navigational chart.

Or

(b) How will you go about describing a chart while ordering a navigational chart?

5. (a) (i) What is the full form of GMDSS?

(ii) What is the basic concept of GMDSS?

Or

(b) What is DIGITAL SELECTIVE CALLING (DSC)? Explain five purposes and what are the frequencies used?

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CP–8005

Sub. Code
34

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAVAL ARCHITECTURE – I

(Upto 2015 Batch)

Time : 3 Hours Maximum : 100 Marks

Answer **all** questions.

(5 × 20 = 100)

1. (a) Briefly explain the development of ocean going merchant vessels.

   (b) Draw and explain plan, profile and cross sectional view of a bulk carrier.

   Or

   (c) Define:
   - (i) List
   - (ii) Camber
   - (iii) Freeboard
   - (iv) LWL
   - (v) Draft.

   (d) Draw midship section of a general cargo vessel.

2. (a) A box shaped vessel 100m long and 14m wide is floating in SW at a draft of 7.6 m. The light draft is 3.6 m and load draft 8.5 m. Find the present displacement, DWT available and DWT aboard.

   (b) A box shaped vessel 16*5*12m floats is SW at a draft of 6.1m. Find the RB%.

   Or

(c) Draw plimsol line.

(d) Define :

(i) TPC

(ii) Reserve buoyancy

(iii) Density

(iv) FWA

(v) Archimedes principle.

3. (a) Draw and explain:

(i) Double bottom

(ii) Forecastle deck.

Or

(b) Draw and explain

(i) Peak tanks

(ii) Types of Keel.

4. (a) Explain three states of equilibrium with sketches.

(b) In a vessel of 8800 tonnes displacement and KG 6.2 m, 200 tonnes of cargo was loaded in the LH, 1.7 m above the keel. Find the final KG.

Or

(c) 500 tonnes of cargo was shifted 15 meters vertically down wards in a vessel of 10000 t displacement. Find the effect it has on the ship's KG and state whether KG increase or decreases.

(d) On a ship of 4,950 t displacement, KG 9.2 m, the ship's jumbo derrick is used to load a weight of 50 t, from the wharf, on to the UD (KG 8 m). If the head of the derrick is 25 m above the keel, calculate the KG of the ship.

(i) When the weight is hanging by the derrick on the centre line but 2m above UD and

(ii) Cyter loading.

5. (a) A box shaped vessel 24 × 5 × 3m had an even keel draft of 1.2 m in DW of RD 1.009. Calculate the draft in DW of RD 1.019.

(b) Write differences between stiff ship and tender slip.

Or

(c) A ship of 4000 t displacement has KG 5.1 m, KB 2.1m, KM 5.5m. Find the moment of statical stability when she heels 24; assuming that she is wall-sided.

(d) Define :

(i) free surface effect

(ii) free surface correction

(iii) free surface moment.

CP–8006

Sub. Code
35

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

MARINE ENGINEERING AND CONTROL SYSTEMS – I

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions. (5 × 15 = 75)

1. (a) Write down Newton's three laws of motion.

(b) Communication satellites orbit the earth at a height of 36,000 km. How far is this from the centre of the earth? Is such a satellite has a mass of 250 kG, what is the force of attraction on it from the earth.

Or

(c) Explain what is shear force and bending moments.

(d) Define :

(i) Hardness

(ii) Ductility

(iii) Malleability.

(e) Explain :

(i) Composition of force

(ii) Shear force.

2. (a) Write down the gas law and its equation.

(b) Explain :

(i) Joules law

(ii) Dryness fraction.

(c) An electric heater having resistance of 50 ohms is connected to electric source. If it produces 180 joules of heat in one second, find the potential difference across the electrical heater.

Or

(d) Define :

(i) Sensible heat

(ii) Latent heat.

(e) Explain boiling point and effect of pressure on it.

3. (a) Explain :

(i) Kinetic energy

(ii) Potential energy.

(b) A 14300 kG aeroplane is flying at an altitude of 497 m at a speed of 214 km/H. Determine the aeroplanes total mechanical energy.

Or

(c) Write down the first and second law of thermodynamics.

(d) Define :

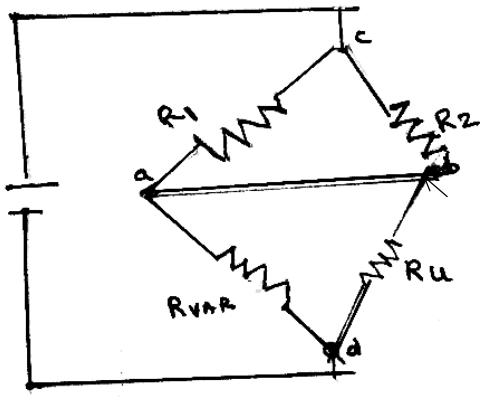
(i) Boyles law

(ii) Charles law.

(e) State and prove conservation law of energy.

4. (a) Explain :
   (i) Electric current
   (ii) Voltage
   (iii) EMF.

(b) A circuit consists of two resistors with resistances R1 = 6.00 ohms and R2 = 1.50 ohms; A variable resistor, the resistance Rvar which can be adjusted, a resistor of unknown value Ru, and 9.0 volt battery connected as shown in the figure. When Rvar is adjusted to 12 ohms, there is zero current through the ammeter what is the unknown resistance Ru?



Or

(c) Define :
   (i) Direct current
   (ii) Alternation and current.

(d) Explain :
   (i) Ohms law
   (ii) Kirchoff's law.

(e) Explain :
   (i) Electrostatics
   (ii) Electro-magnetism
   (iii) Electricity.

5. (a) What is a centrifuge and what is it used for?

(b) Explain how electricity is produced and distributed in a ship.

(c) What are the safety precautions to be taken during bunkering (fuelling) of a ship?

Or

(d) Explain how fresh water is produced from sea water in a ship.

(e) Why is the oily bilge water such an important item in a ships engine room?

(f) What are the various fire fighting fixed and portable equipment available in a ship?

CP–8007

Sub. Code
36

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

ENVIRONMENTAL STUDIES

(Upto 2015 batch)

Time : 3 Hours Maximum : 100 Marks

Answer **all** questions. (5 × 20 = 100)

1. (a) Environmental science is very important to human life-discuss. (10)

   (b) What are the steps to take for protecting our environment? Explain in detail. (10)

   Or

   (c) Is forest boon (or) curse? Discuss. (10)

   (d) Describe the problems associated with the over exploitation of mineral resources and ground water. (10)

2. (a) Discuss the major features of grassland ecosystem. (10)

   (b) What is meant by ecological succession and explain the types of ecological succession? (10)

   Or

   (c) Explain the features of aquatic ecosystem. (10)

   (d) Discuss the structure and functions of ocean eco-system. (10)

3. (a) What is meant by biodiversity? And explain the classification of bio-diversity? (10)

(b) Discuss the status of India as a mega diverse nation of a bio-diversity. (10)

Or

(c) Explain the conservation and strategy of bio-diversity. (10)

(d) What are the various hotspots of bio-diversity in India? (10)

4. (a) What are the reasons behind depletion of ozone layer in the atmosphere? And explain the ill-effects of ozone depletion. (10)

(b) What is meant by acid rain? How does it form? What are the major impacts of it? (10)

Or

(c) How would you control of waste proposal in marine pollution – explain? (10)

(d) List out the various ways in which an individual can contribute towards pollution prevention in the society. (10)

5. (a) What is rain water harvesting? Why it is necessary now-a-days? Explain. (10)

(b) How should public aware in the matter of environmental activities – Explain in detail. (10)

Or

(c) What is AIDS? What are the sources and modes of transmission of HIV infection? (10)

(d) What are the objectives and elements of value education? How can the same be achieved? (10)

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CP–8008

Sub. Code
41

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAVIGATION – IV

(upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) On 13th Sep 2008 in DR 23°21's 047°18'W the observed azimuth of the sun was 046°(c) when the chron showed 01^H 08^M 10^S. If chron error was 02^M 12^S slow and variation was 3°W. Find the deviation of the compass?

Or

(b) On 23rd Sep 2008 in DR 23°40'N 161°56'E the sextant meridian altitude of the suns LL was 66°10.6'. If IE was 2.3' on the arc and HE was 10.5 m. Find the latitude at meriodinal passage and the LOP.

2. (a) On 30th April 2008, in DR 00°20'N 60°12'W the sextant altitude of Suns UL east of meridian was 44°13.4'; when GPS clock showed 13^H 00^M 52^S. If IE was 3.1' off the arc and HE was 20 m. Find the intercept and the direction of LOP.

Or

(b) On 29th Nov 2008 in DR 26°27'N 130°27'W the sextant altitude of suns UL east of meridian was 28°11' when chron showed time 05^H 45^M 44^S chron error 02^M 05^S slow. If HE was 10 m and IE was 2.3' off the arc. Calculate the LOP and longitude where it cuts DR latitude?

3. (a) On 04th March 2008, in DR 27°18'N 168°11'W the sextant altitude of suns LL near the meridian was 56°19.8', when chron showed 11^H 13^M 24^S. If chron error was 01^M 20^S slow, HE was 12 m IE was 2.8' on the arc. Find the direction of the LOP and position through which it passes?

Or

(b) On 01st Sep 2008, AM at strip in DR 18°00'N 178°11'E, the sextant altitude of the polestar was 18°47.4' at 05^H 21^M 08^S by chronometer (error 01^M 18^S slow). HE was 12.5 m, IE was 1.6' on the arc, find the direction of LOP, a position through which it is drawn? If the azimuth was 001°C and variation was 1.3°E. Find the deviation for compass?

4. (a) Write short notes on :

(i) Properties of free gyroscope?

(ii) Errors of a gyro compass?

Or

(b) Write short notes on :

(i) Doppler log

(ii) Navtex.

5. (a) Write short notes on :

(i) LAT

(ii) Chart datum

(iii) Tidal height

(iv) Current

(v) Mean low water.

Or

(b) Write short notes on :

(i) Spring tide

(ii) Neap tide

(iii) Tidal range

(iv) Standard port

(v) Admiralty tide tables.

CP–8009

Sub. Code
42

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

SHIP OPERATION TECHNOLOGY – IV

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Attempt **all** questions. (5 × 15 = 75)

1. (a) What are the types of cargo pipe line? (3)

   (b) What is free flow system of cargo pipe line? Draw free flow system pipe line? (12)

   Or

   (c) What are the safety procedures in a Tanker when cargo operations? Explain in detail. (15)

2. (a) Define Container Carrier. (4)

   (b) What are the types of containers? (4)

   (c) What are the features to construct an ISO standard of containers. (7)

   Or

   (d) Define the following.

      (i) Tanktainer (6)

      (ii) Bulktainer

      (iii) Refrigerated.

(e) What are the container markings and name the important parts of a container? (7)

(f) What is IMDG code? (2)

3. (a) What are the different types of Dry and liquid bulk cargoes? Explain in brief. (12)

(b) What is bulk cargo? (3)

Or

(c) What is Hygroscopic cargo? (3)

(d) What are the checklist prior loading Hygrographic cargoes? (8)

(e) What do you mean by Cargo Shifting bulk cargo. (4)

4. (a) Define the following anchoring terms. (5)

(i) Up-and-down

(ii) Foul Anchor

(iii) Clearing Anchor

(iv) Clear Hawse

(v) Open Hawse.

(b) What are the procedures for maintenance of anchor and cable? (6)

(c) What is abandon ship? Who will declare as abandon ship when vessel in distress? (4)

Or

(d) Describe the actions to take as an OOW when Versel aground. (8)

(e) What are the steps to take when a message received from a Distress Vessel? (7)

5. (a) What are the different types of paint defects? Explain them. (5)

(b) What are the causes for paint defects and corrective active to prevent paint defects? (10)

Or

(c) What are the procedures for chipping and preparation for metal surface painting? (10)

(d) List the common types of corrosion. (2)

(e) What is the Cathode Protection? (3)

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CP–8010

Sub. Code
43

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

VOYAGE PLANNING AND COLLISION PREVENTION AND MARINE COMMUNICATION – IV

(Upto 2015 batch)

Time : 2 Hours Maximum : 25 Marks

Answer **all** questions.

(5 × 5 = 25)

1. (a) Give a detailed note on "Planning and Executing a coastal passage".

Or

(b) Describe the various problems involved in Selection of Ocean Routes.

2. (a) Enumerate the activities involved in Radar Plotting Exercises.

Or

(b) "Revision of Radar Plotting Syllabus has done in second year" — Discuss in detail.

3. (a) While steering 076° (T) at 13 knots, a lighthouse bore 046° (T) 4.5 M off at 0800. At 0809, it bore 029° (T) 2.5 M off. Find :

(i) CPA range, bearing and time

(ii) Beam range, bearing and time

(iii) Set and rate of current

Or

(b) While steering 005° (T) at 18 knots, the following observation were made on the radar screen

1300 hrs .... target bearing 345° range 10 M
1306 hrs .... target bearing 342° range 8.1 M
1312 hrs .... target bearing 338° range 6.4 M

Find :

(i) Set and rate of current

(ii) CPA range and time

(iii) Course, speed and aspect (at 1312) of moving target

4. (a) Explain Annex IV of Collision Regulation in detail.

Or

(b) Explain Rule No. 8 and Rule No. 9 of International Collision Rules.

5. (a) Write short notes on navigational warnings.

Or

(b) What are the contents of Indian Notices to Mariners?

CP–8011

Sub. Code
44

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

MARINE ENGINEERING AND CONTROL SYSTEMS — II

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) What is effect of carbon presence in steel?

   (b) List various types of heat treatment processes.

   Or

   (c) Briefly describe the how steel is produced.

   (d) For what purpose are ceramics used.

2. (a) List major auxiliaries in a ships engine room.

   (b) Where is compressed air used in a ships engine room?

   Or

   (c) Describe the working of a evaporation type fresh water generator.

   (d) How does a hydrofore for sanitary water work?

3. (a) With the aid of a line sketch describe the functioning of a vapour compression refrigeration system. Also mention the state of the refrigerant at various stages.

(b) List various types of pumps used in a ships engine room. Also mention what type of pumps are used for pumping fuel oil, lubricating oil and fresh water and sea water.

Or

(c) Briefly describe parts of a electro hydraulic type of steering gear.

(d) What are the emergency arrangements provided in case of steering gear failure?

4. (a) How are internal combustion engines broadly classified?

(b) Describe the four strokes of a four stroke engine and the processes taking place in each stroke.

Or

(c) What is dual fuel propulsion engine? What is its relevance regarding pollution prevention?

(d) What is exhaust gas turbo charging? What are its advantages?

5. (a) Describe how voltages are stepped up or stepped down. Briefly describe the device which achieves this information.

(b) What is parallel running of generators?

Or

(c) What are the protection devices provided in the electric system and components?

(d) Describe the function of a transformer. How is voltage stepped up or stepped down?

CP–8012

Sub. Code
45

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAVAL ARCHITECTURE – II

(Upto 2015 batch)

Time : 3 Hours Maximum : 100 Marks

Answer any **five** questions.

(5 × 20 = 100)

1. (a) Describe with the main construction features of a watertight bulkhead. (10)

   (b) What is the purpose of subdivision of ships? (2)

   (c) List the functions of : (2 × 4 = 8)

   (i) Transverse bulkheads

   (ii) Longitudinal bulkheads.

2. (a) What is Pounding? (4)

   (b) What is Panting? (4)

   (c) Derive the formula. (3 × 4 = 12)

   (i) FWA

   (ii) TPC

   (iii) BM(T)

3. Answer the following : (5 × 4 = 20)

   (a) Pump rooms

   (b) Galley

   (c) Casting

   (d) Forging

   (e) Continuity of strength.

4. (a) M.V HINDSHIP at a draft of F 5.38m, A 6.17m, GM (Fluid) 0.83m FSC 0.092m

Discharges 430t from No.3 TD, Vcg 10.2m and loads 250t in No.5 LTD. 300t of fuel oil was received equally distributed in No.2 DB tanks (P&S). Calculate her GM(Fluid). (15)

(b) A box shaped vessel 210 m long, 25m wide floats in SW at a draft of 10m F & A, KG = 10Gm. An empty mid ship compartment 30m × 25m has a W/T flat 10.Sm above the keel, find the GM?

5. (a) A ship of 2000 tonnes displacement and KG 3.66m, loads 1500 tonnes (KG 5.5 m) 3500 tonnes (KG 4.60 m) and takes 1520 tonnes (KG 0.6 m). She discharges 2000 tonnes cargo (KG 2.44 m) and consumes 900 tonnes of bunkers (KG 0.40). Find the KG of the vessel at the end of the voyage. (8)

(b) Explain the following : (3 × 4 = 12)

(i) Righting lever and righting moments.

(ii) Timber deck cargo and its effects on stability.

(iii) How to reduce grain heeling moments while loading grain cargo?

6. (a) A box shaped vessel 100 × 20 × 8 m floats in SW upright at 5m draft. Find the new draft if the vessel list 16 deg?

(b) Will a homogeneous log 6m × 3m × 3m and relative density 0.4 float in Fresh water with a side perpendicular to the waterline. If not, what will be the angle of Loll?

CP–8013

Sub. Code
46

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

OCEANOGRAPHY AND METEOROLOGY – II

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) Describe about the formation of general circulation of atmospheric with suitable diagram.

Or

(b) Describe the concepts of Divergence and Convergence in metrology.

2. (a) Define the following concepts

(i) Pressure Gradient Forces

(ii) Coriolis Force

(iii) Geostrophic wind.

Or

(b) Describe about Geostrophic wind scale and how Geostrophic wind approximations are obtained from charts.

3. (a) Describe the dynamic mechanism of Indian Monsoon.

Or

(b) Describe the following

(i) Land and Sea Breeze

(ii) Anabatic and Katabatic winds.

4. (a) Describe the major factors influence in the formation of ocean circulation system.

Or

(b) Give a short note on major ocean currents of Atlantic Ocean.

5. (a) Explain the structure and types of ocean waves? What are the behavioural characteristics of waves in deep and shallow water?

Or

(b) Define Tide. Explain the formation and types of tides.

CP–8014

Sub. Code
51

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

COMPUTER PROGRAMMING AND UTILITIES

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions. (5 × 15 = 75)

1. (a) Draw the block diagram of a digital computer and explain.

Or

(b) Describe the different types of programming languages. Also compare them.

2. (a) What are the characteristics of database? Also explain the need for database.

Or

(b) How can you add, remove and change information with SQL? Explain with examples.

3. (a) What are the loop structures available in C? Explain them.

Or

(b) Write a program in C to arrange the given set of numbers in ascending order.

4. (a) List the activities that you can do on the internet and explain.

Or

(b) Draw the OSI reference model and explain each layer.

5. (a) How can you manipulate cell contents? Explain.

Or

(b) Write the procedure for creating a graph.

CP–8015

Sub. Code
52

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAVIGATION — V

(Upto 2015 Batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) Explain with diagram the principles of working of a marine radar.

   (b) What is ARPA? Explain the uses of ARPA as an equipment on a ship.

   Or

   (c) Describe the characteristics and limitations of a marine RADAR.

   (d) What all entries are made on a RADAR log book?

2. (a) What is a course recorder? Define the working principle of a course recorder.

   (b) What are the inputs given to a course recorder?

   Or

   (c) Define working principle of an autopilot.

   (d) What are the controls and limitations of an autopilot?

3. (a) Write short notes on :
   (i) Rain clutter
   (ii) Sea clutter
   (iii) Follow up steering
   (iv) Non follow up steering
   (v) Steering motor alarms.

Or

   (b) Write short notes on :
   (i) Spoking
   (ii) Multiple echoes
   (iii) AIS
   (iv) VDR
   (v) Smoke detector alarms.

4. (a) Draw and explain navigation lights alarm system on bridge.

Or

   (b) Explain with diagram various types of displays of a marine radar.

5. (a) What is working principle of a gyrocompass?
   (b) What are the errors of a gyrocompass?

Or

   (c) Write short notes on :
   (i) Tank pressure alarms
   (ii) Gyro alarm
   (iii) Vertical beam width
   (iv) Horizontal beam width
   (v) Pulse length.

CP–8016

Sub. Code
53

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

VOYAGE PLANNING AND COLLISION PREVENTION AND MARINE COMMUNICATION – V

(Upto 2015 Batch)

Time : 2 Hours Maximum : 25 Marks

Answer **all** questions.

$(5 \times 5 = 25)$

1. (a) (i) What precautions are required by the Rules for vessels nearing a bend in a channel where approaching vessel may not be clear?

(ii) What are the lights required for rowing boats whether under oar or sail?

(iii) What are the Rule governing narrow channels?

(iv) How can risk of collision be determined?

Or

(b) (i) Does the fact of your having or not having way upon your ship have any bearing on whether you are "under way" according to the Rules?

(ii) At what point does a vessel cease to be privileged in a crossing or overtaking situation?

(iii) In a crossing situation, assuming that your vessel and the other steamer both hold course and speed, state how you could make use of the range of his masthead and after range light to determine:

(1) Whether he will cross ahead of you

(2) Whether you will cross ahead of him

(3) Whether risk of collision exists

(iv) What are the requirements of the Rules with respect to anchor lights for a vessel 50 meters or upward in length?

2. (a) Describe the various components of Voyage Planning.

Or

(b) Give the detailed note on the following :

(i) Limit of Restricted Area

(ii) Ebb Stream of 3 knots

3. (a) How can risk of collision be determined? Elucidate the Rule governing Narrow Channels.

Or

(b) Give a detailed note on "Annexes of International Regulations for Prevention of Collision and its necessities'.

4. (a) What is GMDSS and what are the equipments used in GMDSS?

Or

(b) What is WWNS?

5. (a) What is DSC?

(b) What is weather Routing?

Or

(c) Define the following:

(i) Safety Message

(ii) Distress Message

(iii) Emergency Message.

CP–8017

Sub. Code
54

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017.

Nautical Science

SHIP MASTER'S BUSINESS

(Upto 2015 batch)

Time : 3 Hours Maximum : 100 Marks

Answer **all** questions.

(5 × 20 = 100)

1. (a) Write in detail the role of master with respect to employer, Cargo owner, employees and environmental protection. (20)

Or

(b) Discuss the contribution of STCW convention to the safety of shipping industry. (10)

(c) Shipping master is an important element — Discuss. (10)

2. (a) What is house keeping and how to ensure hygiene in the ship? (10)

(b) What different entries are made in the OLB and what is its sanctity? (10)

Or

(c) What is the customs house procedure with respect to crew? (10)

(d) What is the role of PHO? (10)

3. (a) List the reports which should be rendered by the master to various authorities such as Government, Customs, Owners, etc. (20)

Or

(b) What are the salient features of capt/pilot relationship? (15)

(c) Write short notes on salvage. (5)

4. (a) What are the salient features of the convention on limitation of liabilities for maritime claims 1976? (15)

(b) Define general average. (5)

Or

(c) Define marine salvage. (5)

(d) What are the points to be taken into account prior to offering salvage services and while accepting? (15)

5. (a) What are the salient features of Bareboat chartering and time chartering? (20)

Or

(b) What is the role of classification societies and what are their duties and responsibilities? (12)

(c) Differentiate between flag state and port state controls. (8)

CP–8018

Sub. Code
55

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

MARINE MANAGEMENT AND COMMERCE

(upto 2015 batch)

Time : 3 Hours Maximum : 100 Marks

Answer **all** questions.

(5 × 20 = 100)

1. (a) Define communication and elaborate different types of communication. (10)

   (b) What are the internal and external barriers to good communication? (10)

   Or

   (c) Behaviour of every individual affects an organisation in its own way – Discuss. (10)

   (d) Developing inter-personal relationship in marine environment is demanding – Discuss. (10)

2. (a) Why is there a recession in the shipping industries? What are your suggestions to overcome recession? (10)

   (b) Do you see future for unmanned ships? Give reasons. (10)

   Or

   (c) How is the Indian shipping organised? (20)

   [to cover administration, training, surveys and ports]

3. (a) What do you mean by maritime fraud? What measures are enforced to check this menace? (10)

(b) What are different documents presented to the customs department while entering and leaving harbour? (10)

Or

(c) What is the role of customs house in eliminating maritime fraud? (10)

(d) Trace the development of Indian ports [both major and minor]. (10)

4. (a) Explain the role of (20)

(i) Charterer

(ii) Ship broker

(iii) Agent

(iv) Ship chandlers

(v) Clearing agents.

Or

(b) With the help of a line diagram explain the functioning of a shipping company. (15)

(c) Explain the functions of DPA. (5)

5. (a) Mention any five types of chartering and explain.(20)

Or

(b) Differentiate between mates receipts and bills of lading. (10)

(c) Mention any five different types of bills of lading and elaborate. (10)

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CP–8019

Sub. Code
61

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

MARITIME LAW

(Upto 2015 batch)

Time : 3 Hours Maximum : 100 Marks

Answer **all** questions. (5 × 20 = 100)

1. (a) Discuss about the structure of IMO and primary purpose of IMO in the shipping industry. (20)

Or

(b) Write short note on UNCLOS. (10)

(c) Describe zones of jurisdiction established by UNCLOS. (10)

2. (a) What are the different classes of contract?

(b) What are the essentials for a contract to be valid? (20)

Or

(c) What are the different flag types w.r.t getting nationality to ships? (10)

(d) What is FOC? What are the criticism in the shipping industry about flags of convenience? (10)

3. (a) Explain the procedure for registering a second hand ship in INDIA. (10)

(b) State briefly what responsibilities rest on the employer of a seaman who is left behind abroad. (10)

Or

(c) What are the mandatory certificates to be carried on board a ship? (10)

(d) Under what circumstances is a ship deemed to be unsafe and thereby subject to detention? (10)

4. (a) What is a charter party? (5)

(b) Write short notes on :

(i) Voyage charter party

(ii) Time charter party

(iii) Demise (bare boat) charter party? (15)

Or

(c) Explain fully the meaning of the terms "Demurrage", "Despatch" and "Dead freight" w.r.t to charter parties. (10)

(d) What are the essential functions of a bill of hading? Write short notes. (10)

5. (a) Explain the role of P and I club. (10)

(b) What is the role of seafarer when an accident/serious injury/dangerous incident occurs on board? (5)

(c) Discuss the information which needed to be reported in an incident/accident report forms? (5)

Or

(d) Write short notes on layday, laytime. (5)

(e) Within C/P terms when will you consider a vessel as "arrived ship"? What is NOR? When should a master tender NOR? (15)

CP–8020

Sub. Code
62

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

NAVIGATION – VI

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions. (5 × 15 = 75)

1. (a) Explain with the help of diagram the working principles of a marine radar.

   (b) What are the limitations of a marine radar?

   Or

   (c) Describe the procedure for switching on/off of a radar and explain the controls available on the radar.

2. (a) Compare between Racon and Ramark? (10)

   (b) Why we should not use floating objects for determining the vessels position fixes? (5)

   Or

   (c) Write short notes on :

   (i) SART

   (ii) EPIRB

   (iii) VDR.

3. (a) Explain the working principle of echo sounder with diagram. (10)

(b) What are the errors of an echo sounder? (5)

Or

(c) Write short notes on :

(i) Differential GPS

(ii) Ramark

(iii) SONAR system

(iv) P.Code.

4. (a) Explain the functioning of a GPS and its structure. (10)

(b) What are the errors of a GPS? (5)

Or

(c) If the own ships course and speed were 282°(T) and 6 knots. A light vessel bore 225°(T), 3.0 nm off at 0200 hrs. At 0230 hrs it bore 140°(T), 2.0 nm off. Find

(i) CPA, bearing, range and time?

(ii) Beam bearing, range and time?

(iii) Set and rate of current?

5. (a) Define coefficients 'A', 'B' and 'C' of a magnetic compass. (8)

(b) Explain heeling error and how to correct it on a magnetic compass. (7)

Or

(c) During the course of swinging the magnetic compass through 360° following bearings by observation and calculations were obtained as shown below.

	N	NE	E	SE	S	SW	W	NW
Com. Brg :	069	064	064	065	063	063.5	071	077
True Brg :	075	075.5	076	076.5	077	078	079	079.5

(i) Find variation.

(ii) Find coefficients B and C. (15)

CP–8021

Sub. Code
63

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

CONVENTIONS AND REGULATIONS

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) What is gross tonnage and net tonnage? (10)

   (b) What is the significance of tonnage measurements? (5)

Or

   (c) Draw and explain the load line markings. How is it made use of in vessels voyage planning? (15)

2. (a) What is the full form of ISGOTT? Enumerate the principles and procedures for conducting hot work outside designated hot work area in a strip (noninerted)? (15)

Or

   (b) What is P and A manual? Discuss contents of P and A manual as required by guidelines? (15)

3. (a) How many annexes are there in Marpol 73/78? (5)

   (b) Elaborate the discharge inertia of oil as per marpol annex 1 other than machinery space. (10)

Or

(c) How will you prepare the vesssel for an IOPP survey? (10)

(d) How the outcome of IMO conventions gets implemented in a member state? (5)

4. (a) Discuss salient features of STCW 2010 amendments to STCW convention and about SOLAS conventions. (15)

Or

(b) Define

(i) Protocol

(ii) Amendments

(iii) Tacit acceptance

(iv) ISM

(v) DOC.

5. (a) What is the role and functions of classification societies? (7)

(b) What is IACS? Write short notes on IACS. (8)

Or

(c) List out IMDG classes of dangerous goods. (8)

(d) What is flag state and port state? What are the grounds on which a port state control inspector can detain a vessel? (7)

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CP–8022

Sub. Code
64

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

MARINE ENGINEERING AND CONTROL SYSTEMS – III

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions.

(5 × 15 = 75)

1. (a) In a large two stroke propulsion engine, describe the method of cylinder lubrication adopted. Also state the properties of the Lube oil used for cylinder lubrication. (10)

(b) What treatment the heavy fuel oil under goes prior to its injection in the cylinder? (5)

Or

(c) Describe with a simple sketch, the working of a ships windlass. (10)

(d) Explain the principle of working of a centrifugal pump. What are its advantages and disadvantages when compared with positive displacement pumps? (5)

2. (a) With the aid of a simple sketch describe the functioning and use of an exhaust gas turbo charger. (10)

(b) Describe the process of scavenging in a two stroke main propulsion engine. (5)

Or

(c) Enumerate the reasons for outbreak of a scavenge fire. What steps are to be taken to extinguish the fire? (10)

(d) In a large two stroke engine, how are main, bottom end and cross-head bearings lubricated? (5)

3. (a) By means of a simple sketch explain the functioning of a thrust block. (10)

(b) What is propeller slip and how is it calculated? (5)

Or

(c) By means of a sketch describe the functioning of a two stage oily water separator. What is the PPM at the outlet of each stage? (10)

(d) What are the criteria for safe discharge of ships sewage disposal? (5)

4. (a) Describe the sensors, transmitters and transducers in the automation chain. (10)

(b) How is Remote control of Rudder achieved? (5)

Or

(c) Explain working of a CP Propeller. (10)

(d) What is the use of Bow Thrusters in a ship? (5)

5. (a) Enumerate the precautions to be taken prior to CO_2 flooding of Engine room. (10)

(b) How is Foam generated and distributed? (5)

Or

(c) List all portable hand held extinguishers in a ship. Explain their specific usage and suitability for different classes of fires. (10)

(d) How are fires broadly classified? (5)

CP–8023

Sub. Code
65

B.Sc. DEGREE EXAMINATION, NOVEMBER 2017

Nautical Science

OCEANOGRAPHY AND METEOROLOGY – III

(Upto 2015 batch)

Time : 3 Hours Maximum : 75 Marks

Answer **all** questions. (5 × 15 = 75)

1. (a) Describe the structure and life cycle of a frontal depression. (15)

Or

(b) Give a short note on the following (15)

(i) Occluded front

(ii) Frontogenesis

(iii) Warm front.

2. (a) Describe the weather associated with a tropical revolving storm and the practical rules of navigation to be applied when in the vicinity of a TRS. (15)

Or

(b) Describe the origin, formation movement and structure of a TRS. (15)

3. (a) What do you meant by weather forecasting? Describe the principles of meteorological analysis. (15)

Or

(b) Give a detailed account on voluntary observation fleet of IMD. (15)

4. (a) Describe the factors to be considered for voyage planning and weather routing. (15)

Or

(b) Describe the parameters for optimum routing and ship performance curve. (15)

5. (a) Give a detailed account on atmospheric pollution by marine transportation. (15)

Or

(b) Write a note on MARPOL 1973/78. (15)

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